

MATH120-24S2 Discrete Mathematics

Assignment 5

1. (a) What are the invertible elements of $\mathbb{Z}_{18}$? Find their inverses.
(b) Solve the equation $11x + 3 = 10$ in $\mathbb{Z}_{18}$.

Solution:

- (a) The invertible elements in $\mathbb{Z}_{18}$ are the elements that are coprime to 18. These are:

$$\{1, 5, 7, 11, 13, 17\}.$$

Since $1 \cdot 1 \equiv 1 \pmod{18}$, $1^{-1} = 1$. Since $5 \cdot 11 \equiv 1 \pmod{18}$ we see that $5^{-1} = 11$ and $11^{-1} = 5$. Similarly, $7 \cdot 13 \equiv 1 \pmod{18}$ so $7^{-1} = 13$ and $13^{-1} = 7$. Finally, $17 \cdot 17 \equiv 1 \pmod{18}$, so $17^{-1} = 17$.

- (b)

$$11x + 3 = 10$$

$$11x = 7$$

$$x = 11^{-1} \cdot 7$$

So we need to find the inverse of 11 in $\mathbb{Z}_{18}$. You should have already found this in part (a) but if not, below is the method.

Now

$$18 = 1 \cdot 11 + 7$$

$$11 = 1 \cdot 7 + 4$$

$$7 = 1 \cdot 4 + 3$$

$$4 = 1 \cdot 3 + 1$$

Retracing our steps we find that $1 = 4 - 3 = 4 - (7 - 4) = 2 \cdot 4 - 7 = 2 \cdot (11 - 7) - 7 = 2 \cdot 11 - 3 \cdot 7 = 2 \cdot 11 - 3 \cdot (18 - 11)$, and hence,

$$1 = 5 \cdot 11 - 3 \cdot 18.$$

This means that $1 \equiv 5 \cdot 11 \pmod{18}$, so 5 is the inverse of 11 in $\mathbb{Z}_{18}$. We find that $x = 11^{-1} \cdot 7 = 5 \cdot 7 = 35 = 17$.

2. Using Fermat's Little Theorem, find the remainder when 5^{1702} is divided by 47.

Solution:

We know that 47 is prime, so $5^{46} \equiv 1 \pmod{47}$ by Fermat's little theorem.

$$5^{1702} = 5^{37 \cdot 46} = (5^{46})^{37} \equiv 1^{37} \cdot 1 \pmod{47}.$$

So the remainder when dividing 5^{1702} by 47 is 1.

3. A *Sophie Germain prime* is a prime p such that $2p + 1$ is prime too. The prime $2p + 1$ is then called a *safe prime*.

A *Mersenne prime* is a prime of the form $2^n - 1$ for some integer $n \geq 2$. It can be shown that if $2^n - 1$ is a Mersenne prime, then n is prime.

We asked ChatGPT for a medium-sized Sophie Germain prime.

Certainly! A medium-sized Sophie Germain prime could be $p = 383$.

For $p = 383$, $2p + 1 = 767 + 1 = 768$, which is not prime. However, $p = 383$ itself is prime, satisfying the definition of a Sophie Germain prime.

We: Isn't $2p + 1 = 767$?

You're absolutely correct, my apologies for the oversight. Let's correct that:

For $p = 383$, $2p + 1 = 2(383) + 1 = 766 + 1 = 767$, which indeed is not prime. However, $p = 383$ itself is prime, making it a Sophie Germain prime. Thanks for catching that mistake!

We: But shouldn't 767 be prime as well in order for $p=383$ to be a Sophie Germain prime?

You're absolutely right. I apologize for the oversight. For $p = 383$ to be a Sophie Germain prime, both p and $2p + 1$ must be prime. Let's correct that:

For $p = 383$, $2p + 1 = 2(383) + 1 = 766 + 1 = 767$.

However, 767 is not prime, which means $p = 383$ doesn't satisfy the definition of a Sophie Germain prime.

Let me correct the example. Let's choose a different p such that both p and $2p + 1$ are prime. Let's consider $p = 47$:

For $p = 47$, $2p + 1 = 2(47) + 1 = 94 + 1 = 95$.

Now, let's verify if both p and $2p + 1$ are prime:

- $p = 47$ is prime.

- $2p + 1 = 95$ is prime.

So, $p = 47$ is indeed a Sophie Germain prime. Thank you for catching that error!

- Explain why ChatGPT is still wrong
- Show that 53 is a Sophie Germain prime. Use trial division to show that the numbers you claim to be prime, are indeed prime numbers.
- Show that if q is a Mersenne prime with $q > 7$ then q is not a safe prime.

Solution:

- 95 is not a prime number, since it is divisible by 5.
- We claim that 53 and $2 \cdot 53 + 1 = 107$ are both prime numbers. We look at 53 first and will show that all prime numbers smaller than $\sqrt{53}$ do not divide 53. Hence need to check 2, 3, 5, 7. We easily find that $53 = 2 \cdot 26 + 1$, $53 = 3 \cdot 17 + 2$, $53 = 10 \cdot 5 + 3$, and $53 = 7 \cdot 7 + 4$, so none of 2, 3, 5, 7 is a divisor of 53. Similarly, since 7 is the largest prime number smaller than $\sqrt{107}$, we only need to check whether any of 2, 3, 5, 7 divides 107. Again, we easily find that $107 = 2 \cdot 53 + 1$, $107 = 3 \cdot 35 + 2$, $107 = 21 \cdot 5 + 2$ and $107 = 15 \cdot 7 + 2$, so none of 2, 3, 4, 7 divides 107.
- Suppose that q is a Mersenne prime and q is a safe prime. Then $q = 2^n - 1$ for some integer n , where we also know that n is prime. Since $q > 7$, $n > 3$. Since q is a safe prime, we have that $q = 2p + 1$ for some prime p . We find

$$\begin{aligned} 2p + 1 &= 2^n - 1 \\ 2(p + 1) &= 2^n \\ p + 1 &= 2^{n-1} \\ p &= 2^{n-1} - 1 \end{aligned}$$

This means that p is a Mersenne prime as well, which in turn implies that $n - 1$ is a prime. But since n is a prime with $n > 3$, n is odd, so $n - 1$ is even. But since $n - 1 > 2$, this is a contradiction.